

write_floorplan

NAME

write_floorplan
Writes floorplan information to files.

SYNTAX

status write_floorplan
[-include include_list]
[-exclude exclude_list]
[-output dir_name]
[-force]
[-nosplit]
[-format icc | icc2]
[-objects objects_list]
[-blocks blocks_list]
[-def_units 100 | 200 | 1000 | 2000 | 10000 | 20000]
[-compress gzip | none]
[-add_def_dependencies true | false]
[-via_as_fixed]
[-routed_nets]
[-include_physical_status include_physical_status_list]
[-exclude_physical_status exclude_physical_status_list]
[-cell_types cell_types_list]
[-net_types net_types_list]
[-read_def_options option_string]
[-def_version version]
[-include_tech_via_definitions]

Data Types

include_list	list
exclude_list	list
dir_name	string
objects_list	collection
blocks_list	collection
include_physical_status_list	list
exclude_physical_status_list	list
cell_types_list	list
net_types_list	list
option_string	string
version	string

ARGUMENTS

-include include_list
Specifies the types of data to include in the output. This option cannot be specified together with the -exclude or -objects options. If this option is not specified, all types are included by default.

The supported types are:

- o blockages
- o bounds
- o cells
- o corridors
- o die_area
- o edit_groups
- o io_guides
- o macros
- o module_boundaries
- o nets
- o pin_guides
- o pins
- o route_guides

o rows

o tracks

o vias

o scan_chains

o routing_directions

o voltage_areas

o fills

o pg_metal_fills

o routing_rules

o pg_regions

o placement_attractions

o bump_regions

o anchors

Specify `-include macros` to output only macro cells and skip standard cells. Specify `-include cells` to output all cells. If you specify `-include {macros cells}`, the command issues a warning message that macros will be ignored.

`-exclude exclude_list`

Specifies the types of data to exclude from the output. This option cannot be specified with the `-include` option or `-objects` options. If this option is specified, the command writes all types, except the types specified in `exclude_list`. The allowed data types are the same as for the `-include` option.

`-output dir_name`

Specifies the name of the directory in which to write the floorplan files. The default is `./floorplan`. If the directory already exists, the command issues an error message, unless `-force` is also specified.

`-force` Overwrites floorplan files in the `./floorplan` directory, or in the directory specified by the `-output dir_name` option. By default, the command issues an error message and exits if the directory exists.

`-nosplit`

Writes out long lines, even if the line length is greater than 80 columns. Use this option to simplify post-processing of the output constraint files or when comparing the constraint files by using the UNIX `diff` command. If this option is not specified, the command breaks long lines near column 80 and inserts a line continuation character (`\`).

`-format icc | icc2`

Specifies the target tool for which to write the floorplan. Valid values are `icc` and `icc2`. By default, the value is `icc2`.

To take floorplan information to the Design Compiler tool (in topographical mode) for RTL resynthesis with floorplan information such as macro placement), use the `-format icc` option. The **EXAMPLES** section contains an example of this flow.

`-objects objects_list`

Specifies the objects to include in the output. Supported object types are cells, ports, blockages, voltage areas, bounds, edit groups, routing corridors, module boundaries, PG regions, site rows, bump_regions, anchors, metal_areas and site arrays. Objects of other types are ignored.

This option cannot be specified with the `-include`, `-exclude`, or `-blocks` options.

`-blocks blocks_list`

Specifies the blocks to output hierarchically. If this option is not specified, only the top-level block is written. This option cannot be combined with the `-objects` option.

`-def_units 100 | 200 | 1000 | 2000 | 10000 | 20000`

Specifies the number of units per micron in the DEF file. All database length units are scaled accordingly in the output DEF. If not specified, the units per micron is obtained from the technology length precision. Valid values are 100, 200, 1000, 2000, 10000, and 20000. This option is passed to the DEF writer and the behavior is identical to the -units option of the write_def command.

-compress gzip | none

Specifies the compression type to use when writing the .DEF file. By default, no compression is performed.

-add_def_dependencies true | false

Specifies whether to add dependencies (sections not requested) to the .DEF file to make it self-sufficient. For example, if this option is set to true, if nets are in the -include list but vias are not, the VIAS section will be added anyway; if the option is set to false, it will not be added. The default is true.

-via_as_fixed

Indicates that all the via definitions are written out as fixed vias in the output file.

-routed_nets

Indicates that all the nets written out in the output are routed nets, that is, each net has at least one shape associated it.

-include_physical_status include_physical_status_list

Specifies which cells to include in the output based on their physical status. When you use this option, all of the specified cells are included in the generated DEF file. The physical status can be one or more of the following:

- all
- application_fixed
- fixed
- legalize_only
- locked
- placed
- unplaced

-exclude_physical_status exclude_physical_status_list

Specifies which cells to exclude in the output file based on their physical status. When you use this option, all of the specified cell are included in the generated DEF file except for those whose physical status is specified in the exclude_physical_status_list.

-include_physical_status and -exclude_physical_status are mutually exclusive.

-cell_types cell_types_list

Specifies which cells to include in the output DEF file based on their types. When you use this option, all of the specified cells are excluded in the generated file except for those whose cell types is specified in the cell_types_list. These are the specifiable cell types:

- corner
- end_cap
- pad
- flip_chip_pad
- flip_chip_driver
- pad_spacer
- macro
- lib_cell

-net_types net_types_list

Specifies which nets to include in the output DEF file based on their types. When you use this option, all of the specified nets are excluded in the generated file except for those whose net types is specified in the net_types_list. The net types can be one or more of the following:

- clock
- ground
- power
- signal

-read_def_options option_string

Specifies options to include with the read_def command that is

written to the floorplan file.

`-def_version version`
Specifies the DEF syntax version used to create floorplan.def file. Valid values are 5.7 and 5.8. The default is DEF version 5.8.

`-include_tech_via_definitions`
Includes all via definitions in the technology file, including those not used in the design, in the output DEF file. By default, only via definitions present in the design are written out to the DEF VIAS section.

DESCRIPTION

The `write_floorplan` command writes the floorplan.tcl Tcl script and floorplan.def DEF file. The floorplan.def file generally includes the objects that are numerous, as sourcing the corresponding Tcl would be time-consuming. The floorplan.tcl contains the remaining floorplan constructs and a `read_def` command to load the floorplan.def file. To restore the floorplan, source the floorplan.tcl Tcl script.

By default, the floorplan files are written to a directory named `./floorplan`. The destination can be changed by using the `-output` option. The command also writes a third file, `floorplan_compare_data.txt`, that contains information to use when comparing the output with a new design by using the `compare_floorplans` command. If the `-blocks` option is used, the command creates subdirectories with names corresponding to the list of blocks.

The `write_floorplan` command writes out user attributes associated with objects to the Tcl script. In the IC Compiler tool, there is no shape or via object associated with a terminal. In this case, the script sets user attributes from the terminal and shape or via object on the terminal for the IC Compiler tool. If there is a conflict between values of the same attribute, a higher precedence is given to the value set on the terminal.

This command returns 1 on success, 0 otherwise.

EXAMPLES

The following example writes floorplan files to the script directory.

```
prompt> write_floorplan -output script
```

The following example writes only cells and blockages.

```
prompt> write_floorplan -include {cells blockages}
```

The following example writes macros hierarchically for all blocks.

```
prompt> write_floorplan -include {macros} -blocks [get_blocks *.*]
```

The following example writes the selected objects and ignores objects that belong to unsupported types.

```
prompt> write_floorplan -objects [get_selection]
```

The following example writes floorplan information in icc format. This could be used to export the floorplan information to the Design Compiler tool (in topographical mode) where the RTL is resynthesized with this floorplan information. Non-PG nets and non-fixed cells are not needed for re-synthesis and so they can be filtered out of the DEF file by using post-processing.

```
prompt> write_floorplan -format icc
```

The following example writes out the floorplan file for all blocks. To load back floorplan file for each block, one method is source the floorplan.tcl Tcl script, the other method is use `set_constraint_mapping_file floorplan/mapfile` and `load_block_constraints -type FLOORPLAN -all_blocks`.

```
prompt> write_floorplan -output floorplan -blocks [lsort -unique \
[get_attribute [get_cells -hierarchical -filter is_soft_macro] ref_name]
```

The following example writes out the floorplan file and includes the `-add_def_only_objects {cells}` and `-no_incremental` options to the `read_def` command written to the floorplan file.

```
prompt> write_floorplan -output "dirName" -nosplit -verbosity low \
-def_version 5.8 -read_def_options {-add_def_only_objects {cells} -no_incremental}
```

SEE ALSO

[compare_floorplans\(2\)](#)

[read_def\(2\)](#)

[write_def\(2\)](#)

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